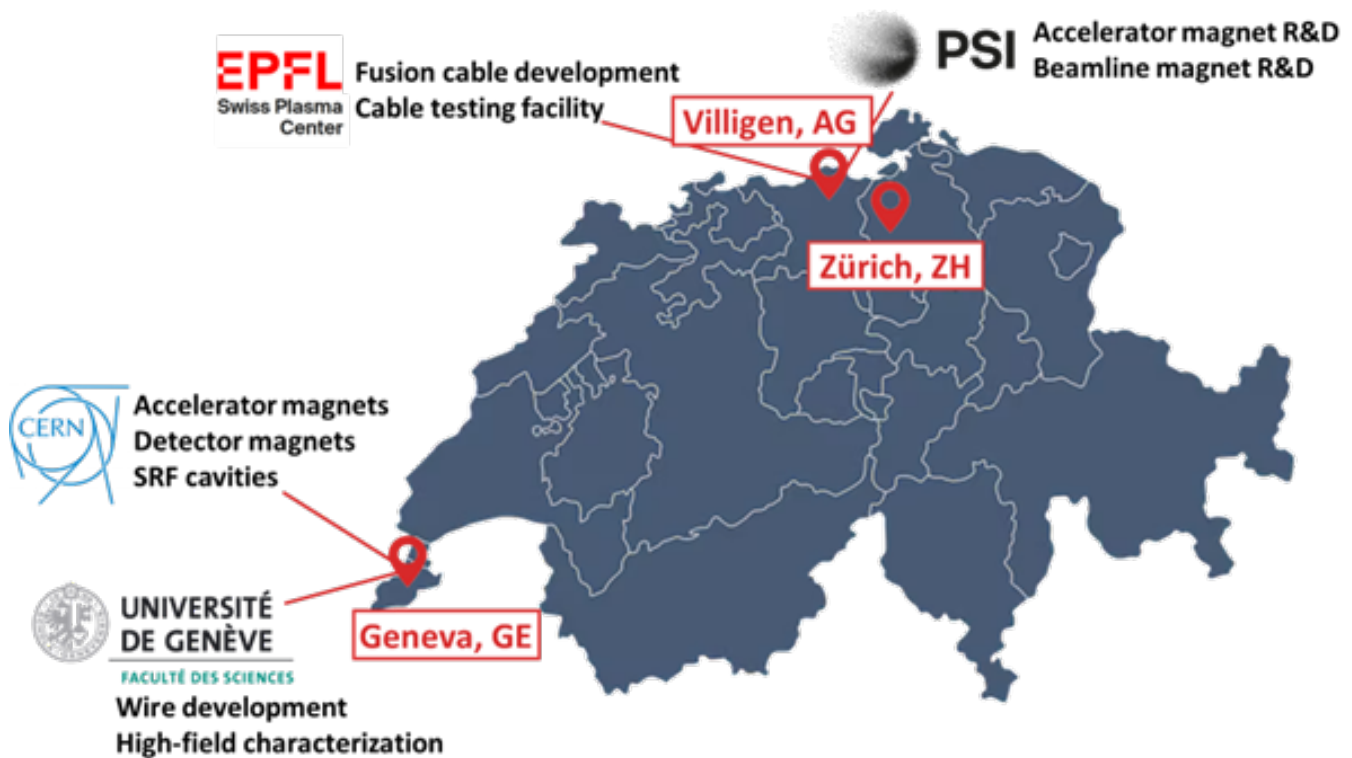


Switzerland:

The state of High Temperature Superconductivity in 2025

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Energy Context Highlights

Switzerland's final energy consumption in 2024 saw a 1% increase to 776,220 TJ, a rise largely attributed to colder weather and an 8.9% surge in aviation fuel sales. Despite this, the energy transition is evident in the heating sector, where ambient heat utilization via heat pumps grew by 6.7% and district heating increased by 9.2%, now maintaining a 43% renewable share. Overall, renewable energy accounts for 29.9% of final consumption, while fossil fuels represent 34.9%. Electricity consumption also rose by 1.4% to 57'500 GWh, making up 26.7% of the energy mix.

Policy, Regulatory, and Infrastructure Highlights

Current energy policy focuses on accelerating the expansion of domestic renewables and ensuring long-term supply security:

- **Regulatory Amendments:** The Federal Council approved changes to the Energy Ordinance (EnV) and Energy Promotion Ordinance (EnFV) to meet 2030 targets, including a winter electricity bonus for photovoltaic (PV) systems.
- **Climate Protection Act (KIG):** This act entered into force, setting clear targets for climate neutrality, supplemented by amendments to the CO₂ Ordinance to bring it into line with EU requirements, and providing financial incentives for technologies such as heat pumps.
- **Supply Security & Nuclear Energy:** Due to concerns regarding long-term supply gaps, the Federal Council proposed relaxing the Nuclear Energy Act to keep existing plants open, a move that has reignited a national debate on technological openness.
- **Expansion Needs:** The SFOE Monitoring Report 2025 indicates that renewable energy expansion (excluding hydropower) must more than double to achieve the mandatory 2035 targets.

R&D Activities Highlights

Swiss HTS research is globally recognized for advancing high-field magnet technologies, driven by future requirements for particle physics and fusion energy:

- **Next-Generation Colliders:** Research is focused on two primary options for CERN's next flagship project after the Large Hadron Collider (LHC) and its high-luminosity upgrade: the Future Circular Collider (FCC), comprising an electron-positron machine (FCC-ee) expected to begin operations in the mid-2040s and a proton-proton collider (FCC-hh) reaching up to eight times the LHC energy, and the Muon Collider, which would collide muon beams to explore new physics with a more compact and potentially sustainable design. For the FCC-hh, studies are comparing 14+ T Nb₃Sn dipoles with 20 T HTS dipoles, the latter potentially enabling collision energies of 120 TeV, roughly eight times that of the LHC. The Muon Collider would rely primarily on HTS magnets, taking advantage of their higher field strength and higher-temperature operation. A critical component of this research is the creation of compact solenoids capable of generating fields beyond 40 T. These high-field magnets are essential for the final cooling of the muon beam, a process designed to reduce transverse emittance to the levels required for effective collisions.
- **High-Fidelity Characterization:** The University of Geneva (UNIGE) is one of the few teams globally capable of performing detailed characterization of commercial REBCO coated conductors. This includes measuring critical current (I_c) dependence on temperature, field intensity, and orientation, as well as a unique experiment probing HTS tape tolerance to transverse tension, which is vital for magnet design optimization.
- **Energy Efficiency in Infrastructures:** The Paul Scherrer Institute (PSI) is developing combined quadrupole–sextupole HTS magnets for the FCC-ee, aiming to achieve a 20% reduction in energy

consumption compared to traditional resistive (normal conducting) designs. This initiative is central to PSI's efforts to lower the overall operating costs of the FCC project.

- **Fusion Conductor Innovation:** The Swiss Plasma Center (SPC) at the École Polytechnique Fédérale de Lausanne (EPFL) is advancing superconducting technologies for nuclear fusion applications through projects like the Volumetric Neutron Source (VNS) and developing innovative HTS conductor concepts for use in plasma confinement magnets, including ASTRA (proposed for the EU DEMO central solenoid) and LASSO (Laminated Stacked-tape Soldered conductor).

Demonstration Projects Highlights

- **Positron-Capture Solenoid:** PSI is leading the development of an HTS solenoid for the FCC-ee injector designed to increase capture efficiency tenfold compared to existing technologies. Commissioning is slated for 2024–2025, with full testing at SwissFEL scheduled for 2025–2026.
- **HTS-based Beamline Solenoids:** PSI is developing a 6 T solenoid for Resonant Inelastic X-ray Scattering (RIXS) at the Swiss Light Source (SLS) and an 18 T split solenoid for the Swiss Spallation Neutron Source (SINQ)
- **Superconducting Joints:** A project funded by the Swiss Innovation Agency between UNIGE and Bruker BioSpin is developing reproducible techniques for superconducting joints in REBCO conductors. Success in this area is critical for the next generation of high-resolution research-grade NMR spectrometers.
- **Fusion Magnet Testing:** The SPC continues to operate the SULTAN facility, conducting critical tests on HTS cables for international projects like the STEP tokamak (UKAEA) and Chinese tokamaks such as BEST and CFETR.

Industrial Players and Activities Highlights

- **Bruker BioSpin:** A world leader in NMR technology, Bruker has developed at its Zürich facility a prototype 1.3 GHz NMR spectrometer featuring a 30.5 T all-superconducting magnet.
- **Fusion Startups:** Germany-based Proxima Fusion has signed a research agreement with PSI and utilizes the SPC for testing HTS conductors.
- **International Collaborations:** Private firms such as Commonwealth Fusion Systems, Tokamak Energy, and Advanced Conductor Technologies actively use SPC infrastructures for testing HTS cable designs.

Success Stories and Lessons Learned

While Switzerland hosts world-class research and testing facilities for HTS in the fields of high-energy physics and fusion, there has been a notable lack of manifested interest from national utilities and the domestic energy industry in adopting HTS-based solutions for heavy industrial or grid infrastructure. This gap between scientific excellence and industrial application in the energy sector is a primary challenge.

Next Term

Switzerland, represented by the Swiss Federal Office of Energy (SFOE), has officially decided to withdraw from the IEA TCP on HTS at the conclusion of the current term. This decision stems from budget constraints and the aforementioned lack of commercial interest from Swiss utilities and energy industries in implementing superconducting technologies. Swiss superconductivity R&D will continue to be driven by specialized scientific institutions (UNIGE, PSI, EPFL) outside the framework of the TCP.